

GDA API Gateway v1 — Build Report

Date: 2026-04-27. Not deployed live. Built per the gateway plan and n8n response standard.

Deliverables

- Source tree: `/home/user/workspace/gda-api-gateway-v1/`
- Zip package: `/home/user/workspace/gda-api-gateway-v1.zip` (~30 KB, excludes `node_modules` and `.env`)

Architecture

Stack: Node 18+ / Express 4 (matches plan Option A). One process. Server-side only. No database, no auth provider yet — Phase 1 is read-only / dry-run.

```
gda-api-gateway-v1/
├── src/
│   ├── server.js           # express app, env loader, CORS, JSON 404/500
│   └── routes/
│       ├── health.js      # GET /health
│       ├── qa.js          # GET /api/qa/health, POST /api/qa/dry-run
│       ├── workflows.js   # GET /api/workflows/registry (?refresh=true → live)
│       └── failures.js    # GET /api/failures/latest
│   └── lib/
│       ├── envelope.js    # ok / fail / notConfigured helpers (matches gda-n8n-res
│       ├── n8nClient.js   # callWebhook + listFailedExecutions + fetchWorkflows
│       └── checks.js      # READONLY_CHECKS, DRYRUN_CHECKS, classify(), recommend
├── data/
│   └── workflow-registry-baseline.json # copy of baseline from /home/user/workspac
├── test/
│   └── smoke.js           # 21 assertions; runs with no secrets
├── package.json
├── .env.example
├── .gitignore
└── README.md
```

Endpoints (all wired, all returning the standard envelope)

Endpoint	Method	Behavior
/health	GET	Liveness. Returns { webhookConfigured, apiConfigured, missing*, uptimeSec, nodeVersion }. Always 200.

Endpoint	Method	Behavior
/api/qa/health	GET	Calls 8 read-only n8n webhooks (same set as React QA Center): gda-platform-health, gda-dashboard-mega, gda-trends, gda-opp-tracker, gda-launchpad-funnel, gda-deep-research-history, gda-capture-plan, gda-morning-briefing. Returns summary, per-row results, plain-English nextAction. If N8N_BASE_URL missing → configured: false envelope.
/api/qa/dry-run	POST	Calls only approved dry-run checks (gda-save-opp, gda-risk with dryRun: true). Server-side re-stamps dryRun: true on every body before sending — clients cannot disable. Optional { ids: [...] } filter; any unapproved id → 400 NOT_ALLOWED.
/api/workflows/registry	GET	Default returns baseline JSON from data/workflow-registry-baseline.json (159 workflows). With ? refresh=true and N8N_API_* set, fetches live from n8n REST.
/api/failures/latest	GET	Calls n8n /executions?status=error&limit=N, maps each to { id, workflowName, failedNode, message, startedAt, stoppedAt }. If env vars missing → configured: false, never crashes.

Response shape

Every endpoint returns the GDA envelope from gda-n8n-response-standard.md:

```
{ "success": true, "workflow": "GDA.gateway.*", "action": "...", "dryRun": false, "
```

When env is missing, the envelope adds configured: false and meta.missing: [...] instead of failing.

Security posture (built in to v1)

- .env ignored by .gitignore. .env.example only contains placeholder keys.
- No secrets ever returned in responses.
- Request bodies are not logged (avoids leaking dry-run inputs).
- x-powered-by disabled.
- CORS allowlist driven by ALLOWED_ORIGINS env, defaults to same-origin.
- Dry-run endpoint enforces dryRun: true server-side after request validation; client cannot bypass.
- Registry route only reads; failures route only reads.

Tests / smoke

`npm run smoke` boots Express on a random port with all env vars unset and runs 21 assertions:

```
[PASS] /health 200
[PASS] /health envelope success:true
[PASS] /health webhookConfigured=false
[PASS] /health apiConfigured=false
[PASS] /api/qa/health 200
[PASS] /api/qa/health configured:false
[PASS] /api/qa/health lists missing N8N_BASE_URL
[PASS] /api/qa/dry-run 200
[PASS] /api/qa/dry-run configured:false
[PASS] /api/qa/dry-run lists allowed ids
[PASS] /api/qa/dry-run rejects non-allowed id with 400
[PASS] /api/qa/dry-run reject error code NOT_ALLOWED
[PASS] /api/workflows/registry 200
[PASS] /api/workflows/registry success:true
[PASS] /api/workflows/registry source baseline-file
[PASS] /api/workflows/registry has workflows array
[PASS] /api/failures/latest 200
[PASS] /api/failures/latest configured:false
[PASS] /api/failures/latest lists missing N8N_API_BASE
[PASS] /api/failures/latest lists missing N8N_API_KEY
[PASS] unknown route returns 404 JSON
```

smoke: OK

This proves the gateway is safe to start with no credentials — it returns `configured:false` envelopes instead of crashing.

How to use

Plain-English Windows + VPS setup is in the package `README.md`. Quick version:

```
unzip gda-api-gateway-v1.zip
cd gda-api-gateway-v1
npm install
cp .env.example .env          # edit with real values when ready
npm run smoke                 # all PASS, no secrets needed
npm start                     # listens on :8787
```

Phase 1 vs Phase 2

This v1 covers exactly the four endpoints listed in the gateway plan as Phase 1. Phase 2 (opportunities/save, risks, capture-plan/preview, deep-research/preview, competitors/crawl-preview) is **not** in v1 — that scope needs the audit log + approval token, which were intentionally deferred per the plan.

Known follow-ups

- Plug `/api/qa/health` into the React QA Center as a single source-of-truth call (replaces the 8 individual fetches the page makes today). Out of scope for the gateway repo itself.
- Decide where the gateway runs in production (alongside React in the same docker compose, or its own service). The package has no opinion — just `npm start` + a port.
- Add an audit-log writer when Phase 2 begins.

Constraints honored

- No live deploy.
- No secrets in source. `.env.example` placeholders only.
- No DB writes. Only n8n calls; the only writes possible are dry-run and they are forced server-side.
- Smoke test runs offline with zero credentials.